

Channabasavanna Santosh Pawate

AI Engineer & Full-Stack Developer

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PROFESSIONAL SUMMARY

AI Engineer with 4+ years of production experience shipping LLM-powered systems, RAG pipelines, and full-stack AI applications. Sole developer across three live platforms at Lakshmi Silk House; deployed a production AI chatbot (Qwen 3.5B + RAG) cutting support response time by 60%; fine-tuned ViT-B/16 achieving 97.6% weighted F1. MSc Artificial Intelligence (Merit), Coventry University 2026. Graduate Visa holder.

CORE COMPETENCIES

Python / FastAPI

LLM Integration (Gemini, Groq, Qwen)

RAG Pipelines / ChromaDB

Prompt Engineering & Fine-Tuning

PyTorch / Computer Vision

React / Next.js / TypeScript

C# / .NET / ASP.NET Core

Docker / Linux VPS

REST API Development

Explainable AI (SHAP)

SQL / MySQL

WORK EXPERIENCE

Software Developer

Jan 2021 – Apr 2025

Lakshmi Silk House – Karnataka, India

- Architected and deployed a production AI chatbot integrating Qwen 3.5B LLM with a RAG pipeline (ChromaDB, FastAPI) and custom prompt engineering on a Linux VPS-hosted Magento e-commerce platform, reducing manual support response time by over 60%.
- Engineered an AI Catalog Generator using Next.js 15, React 19, and Gemini 2.5 Flash Image API to automate Instagram-ready product catalogue creation from raw product photos, cutting asset production time by 70%.
- Maintained and optimised three concurrent production platforms as sole developer – Python REST API, C#/.NET desktop billing system, and Flutter mobile inventory app – all Git-versioned, unit tested, and MySQL-backed, ensuring zero unplanned downtime.
- Refactored Flutter mobile inventory application, achieving a 60% reduction in load time and a 40% increase in daily active users within two weeks of deployment.
- Built and deployed an end-to-end ML return prediction pipeline using Random Forest (85% accuracy), SHAP explainability, and a Streamlit dashboard, reducing return processing errors by 80% and saving 10+ hours of review time weekly.
- Integrated Google Gemini API into Doctor AI Prescriber – a C#/.NET Windows Forms desktop application used in a live ENT clinic – delivering AI-powered medicine suggestions, prescription history, and print/export functionality.
- Developed customer churn prediction models using Python, Scikit-learn, Pandas, and NumPy on large retail datasets, identifying high-risk segments and enabling targeted retention strategies.
- Designed and implemented Python backend systems and MySQL databases from scratch, establishing Git version control workflows and REST API design patterns across all production systems.

Student Ambassador

Aug 2025 – Present

Coventry University – Coventry, UK

- Represent Coventry University at academic events, supporting prospective international students and outreach initiatives.

PROJECTS

Project	Stack	Highlight
COVID-19 X-Ray Detection github.com/appupawate2908/covid-detection-xray	PyTorch, ViT-B/16, FastAPI, React, Attention Rollout XAI	97.6% weighted F1
ReviseAI – Telegram Study Bot github.com/appupawate2908/revise-ai-bot	Python, Groq API, Llama 3.3 70B, PyMuPDF, gTTS	PDF → flashcards, quizzes, 7-day study plans

Project	Stack	Highlight
AI Catalog Generator github.com/appupawate2908/catalog-generator	Next.js 15, React 19, TypeScript, Gemini 2.5 Flash, Tailwind	Raw photos → Instagram-ready catalogue
Doctor AI Prescriber github.com/appupawate2908/ai-doctor-prescriber	C#, .NET 10, Windows Forms, Google Gemini API	Live ENT clinic deployment
SmartHire – Job & Candidate Mgmt	ASP.NET Core 8, Razor Pages, SQL Server, EF Core, ASP.NET Identity	Full-stack platform, role-based auth
FinTrack – Banking Dashboard	Angular 17, TypeScript, Angular Material, REST API	Transaction history, account summaries

EDUCATION

MSc Artificial Intelligence and Human Factors

Coventry University – Coventry, UK

Dissertation: COVID-19 X-ray detection (ViT-B/16), 97.6% weighted F1, Attention Rollout XAI, FastAPI + React.

Bachelor of Engineering - Information Science & Engineering

Visvesvaraya Technological University – India

CERTIFICATIONS

Prompt Engineering for ChatGPT – LinkedIn Learning (Jun 2025)

Tata Cybersecurity Analyst Job Simulation – Forage (Mar 2025)

TECHNICAL SKILLS

Programming, Querying & Scripting: Python (Pandas, NumPy, PyTorch, Scikit-learn, FastAPI, Flask), Java, SQL (Joins, CTEs, Window Functions, Subqueries), JavaScript, TypeScript, C#, Dart, HTML/CSS, Bash

Machine Learning, AI & Statistical Analysis: Scikit-learn, PyTorch, TensorFlow, XGBoost, Hugging Face, Classification, Regression, Clustering, Anomaly Detection, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Computer Vision, Explainable AI (SHAP, Attention Rollout), Time Series, A/B Testing, Cross Validation, Dimensionality Reduction

Generative AI & LLM Engineering: LLM APIs (Gemini, Groq/Llama, Qwen, Anthropic), RAG Pipelines, ChromaDB, Prompt Engineering, Fine-Tuning, Agentic Workflows, Multi-Step LLM Pipelines, Model Context Protocol (MCP), Claude Code, GitHub Copilot

Data Visualisation & Reporting: Streamlit, Matplotlib, Seaborn, Plotly, SHAP Visualisation, Dashboard Development, KPI Design, Advanced Excel

Web, Mobile & Frameworks: ASP.NET Core 8, Razor Pages, Entity Framework, React, Next.js, Angular, Flutter, Tailwind CSS, REST API Design, FastAPI, Flask

Databases & Storage: MySQL, PostgreSQL, SQL Server, SQLite, ChromaDB, Data Modelling, Query Optimisation, Indexing

DevOps, Cloud & Tools: Docker, Kubernetes, CI/CD Pipelines (Jenkins, GitHub Actions), Git, GitHub, Linux VPS, Vercel, Heroku, Postman, Jupyter Notebook, VS Code, Agile/Scrum

Finance & Domain Knowledge: Fixed Income Products (Government & Corporate Bonds, Bond Futures, Interest Rate Swaps), Trade Lifecycle & Settlement, Portfolio Management Operations, Data Reconciliation, Bloomberg, Order Management Systems